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AdOpT-NET0: A technology-focused Python package for the optimization of multi-energy systems

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Summary

AdOpT-NET0 (Advanced Optimization Tool for Networks and Energy Technologies) is a software designed to optimize multi-energy systems via linear or mixed-integer linear programming. Energy system optimization models like AdOpT-NET0 are crucial in shaping the energy and material transition to a net-zero emission future. This transition involves various challenges such as the integration of renewable energy resources, the selection of optimal decarbonization technologies and overarching strategies, and the expansion or rollout of new networks (electricity, hydrogen, CO2). At the same time, the interplay of traditionally separated sectors (e.g., the residential, industrial, and power sectors) becomes increasingly important. The resulting systems are inherently complex and at times non-intuitive; not surprisingly, models to simulate and optimize such complex systems are of paramount importance for a successful transition to a NET-0 society.

AdOpT-NET0 is a comprehensive tool to model and optimize a wide range of multi-energy systems from individual technologies to industrial clusters, regions, or multiple countries. In multi-energy systems, multiple energy and material carriers, conversion and storage technologies, as well as means of transport can interact. These systems are highly complex but also offer synergies to reduce costs and environmental impacts. Table 1 provides an overview of the covered dimensions of AdOpT-NET0, while Figure 1 and 2 show two examples of energy systems that can be modeled with the tool.

Table 1: Dimensions of energy system models and features of AdOpT-NET0

Dimension	Feature of AdOpT-NET0
Spatial resolution	Single node or multiple nodes with network constraints
Temporal resolution	Flexible: 15 min to hours (default: 1h)
Number of carriers or materials	Many different, can be user-defined
Technology model detail	Linear or mixed-integer linear (piece-wise if necessary)
Type of analysis	Operation only, brownfield or greenfield design
Time frame	Year or multiple years
Uncertainty	Deterministic, perfect foresight

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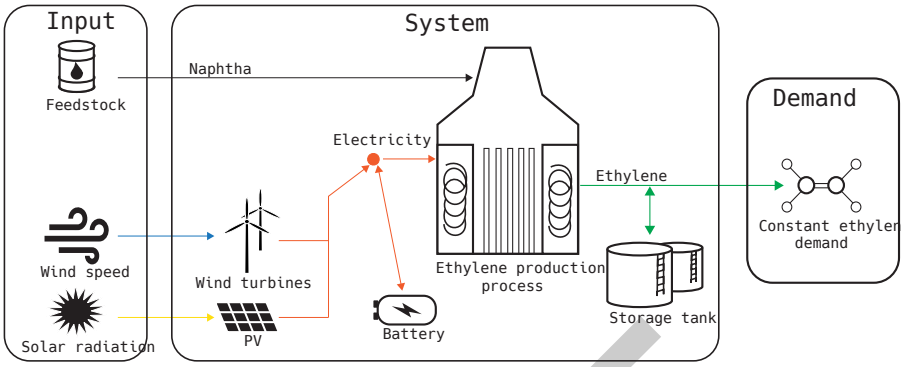

Figure 1: A possible application of AdOpT-NET0 with a single node studying ethylene production with an electric cracker relying on variable renewable energy sources (adapted from Tiggeloven et al., 2023)

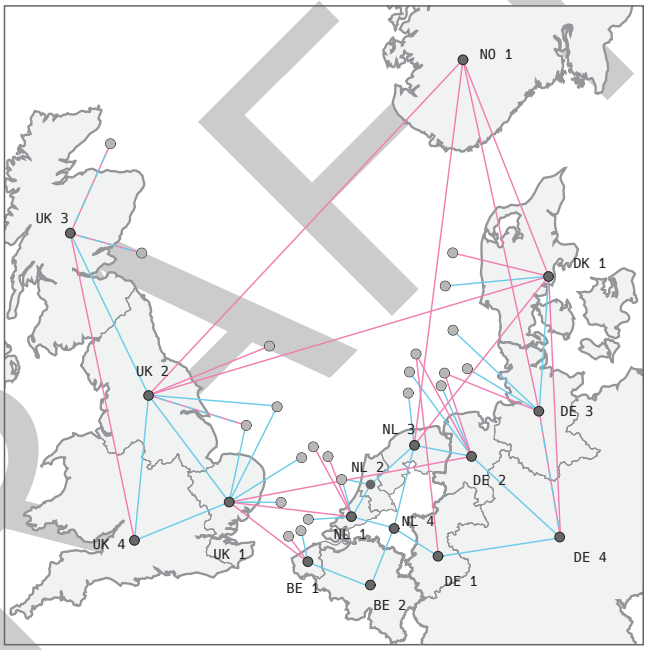

Figure 2: A possible application of AdOpT-NET0 with multiple nodes and networks studying the integration of large-scale offshore wind in the North Sea region (adapted from Wiegner et al., 2024)

The standard formulation of the model framework is a mixed integer linear program. Its mathematical formulation supports flexible spatial and temporal resolution and technological details. AdOpT-NET0 can optimize both system design and technology operation variables, enabling the optimization of existing energy systems with expansions or additions (brownfield) and new systems without the constraints of existing installations (greenfield). A key feature of AdOpT-NET0 is its high level of technological detail which allows for a highly realistic assessment of individual technologies and their integration into an energy system without limiting the scope of the analysis. Furthermore, several complexity reduction algorithms can be adopted to deal with infeasible computation times (Gabrielli et al., 2018; Weimann & Gazzani, 2022).

The tool was developed to assist researchers and students interested in energy system modeling. It combines 5+ years of research and is inspired by a closed-source MATLAB version of the model. Multiple detailed technology models, time aggregation methods, solving heuristics, and general improvements were added to form the present Python package that is further

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developed. AdOpT-NET0 also comes with a web-based visualization platform to provide a quick yet deep understanding of the model results to make informed decisions to advance towards a net-zero future.

Statement of need

Traditionally, models in the energy sector fall into two separate categories: (1) highly complex non-linear process or power system models with limited consideration of inter-temporal dynamics, and (2) low complex mostly linear energy system models with simplified technology performances. AdOpT-NET0 bridges this gap by providing a modeling framework to model the complex behavior of energy and industrial technologies embedded in an energy system. As such, the software comes with a number of detailed technology models that are based on previous scientific work; e.g., for direct air capture and carbon capture (Weimann et al., 2023; Wiegner et al., 2022), heat pumps (Ruhnau et al., 2019; Xu et al., 2022), gas turbine models for different sizes (Weimann et al., 2019), underground hydrogen storage, and electric naphtha cracking (Tiggeloven et al., 2023). Additionally, it includes the possibility to model operational constraints of conversion technologies such as ramping rates, minimum uptime, minimum downtime, the maximum number of start-ups, or standby power (Morales-España et al., 2017).

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